

Project Proposal

“Syncro” is an autonomous Multi-Agent Productivity Suite designed to plan, reason and act with limited human supervision to increase efficiency and productivity of the user.

1. Problem Statement & Motivation

- In the modern day, professionals suffer “admin bloat” and “work about work”. This means that many spend roughly 25% of their workday scheduling meetings, emails, updating project trackers, and status updates. Current tools typically used are passive and only store data rather than act on it.
- The target users include project managers, freelancers, and small business owners juggling multiple workstreams.
- Agentic AI is the solution for this proposal since ambiguity, reasoning, and chaining are required to perform more than just a linear thought process. Interpreting vague requests is needed to make sure that there aren’t any conflicts with the user’s input. The agent must also be able to justify or reason why certain actions or events take priority of the other (gym vs business meeting). Navigating between different tools such as Slack and Google Calendar without human intervention goes beyond standard automation.

2. Use Cases & Scenarios

- Scenario A: The "Project Kickoff" Automation
 - User Intent: "We’re starting the 'Alpha Project' today. Set up the infrastructure."
 - Agent Tasks: The Planner Agent identifies the necessary steps. The Tool Agent creates a new Notion workspace, sets up a Slack channel, and invites team members. The Scheduler identifies a 30-minute kickoff slot that works for everyone involved.
- Scenario B: The "Inbox Zero" Triage
 - User Intent: "Triage my inbox and prepare my afternoon."
 - Agent Tasks: The Summarizer Agent reads unread emails. The Categorizer flags urgent client requests vs. newsletters. The Drafting Agent prepares replies for the user to "Approve/Reject," and the Calendar Agent blocks out "Deep Work" time based on the volume of tasks found in the emails.

3. Initial System Design

- Syncro uses a decentralized multi-agent architecture where agents communicate via a shared "Blackboard" (short-term memory).

Agent Role	Responsibility
The Architect (Manager)	Receives the user's natural language goal and breaks it into a DAG (Directed Acyclic Graph) of tasks.
The Gatekeeper (Security)	Scans proposed actions to ensure no private data is shared externally and verifies "destructive" actions (like deleting a meeting).
The Integrator	Specialized in API calls for Google Workspace, Slack, and Notion.
The Time-Lord	Specialized in temporal reasoning by managing time zones, buffer times, and calendar density.
The Briefing Agent	Synthesizes all actions taken into a concise "End of Day" or "Morning" report for the user.

Tools & Data Sources:

- APIs: Google Calendar API, Gmail API, Notion API, Slack Webhooks.
- Local Data: A local SQLite database to store user preferences (e.g., "I never take meetings before 10 AM").

4. Feasibility & Risks

- Potential Challenges
 - Tool Failures: API downtime or auth token expiration can break the chain. Can be corrected by implementing robust error handling and "human-in-the-loop" prompts when an API returns an error.
 - Privacy: Agents handling email content is a high-sensitivity task. Can be corrected by using local processing for PII (Personally Identifiable Information) scrubbing before sending data to the LLM.
 - Context Window: Managing weeks of calendar data. Can be corrected by only feeding the agent data for the relevant date range.
- Constraints
 - API & Rate Limits: Free-tier or standard API keys have daily caps on "write" actions (creating events or pages).
 - Computation & Performance: Large email threads or project docs can exceed the model's memory window.
 - Reliability & Logic: If the "Planner" makes a mistake, every subsequent agent executes the wrong task.
- Project Timeline
 - Feb 5 - Feb 20: API authentication setup and individual "Tool Agent" development.

- Feb 21 - March 15: Implementation of the Multi-Agent Workflow using a framework like LangGraph for state management.
- March 16 - April 1: User Acceptance Testing with real-world calendar/email data.
- April 2 - April 15: Refinement of the "Gatekeeper" agent and final system evaluation.